

Unsupervised Discovery of Traffic Congestion Patterns from New York City Taxi Trip Records

A pipeline grounded in zone-relative speed deviation, cyclic temporal encoding, and stability-aware K-Means

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Abstract

Urban traffic congestion is a phenomenon that is straightforward to perceive but difficult to characterise quantitatively. Conventional metrics based on absolute travel speed conflate two distinct conditions: zones that are persistently slow due to structural factors such as posted speed limits or roadway geometry, and zones that are transiently slow due to demand-driven congestion. This work proposes a clustering pipeline that addresses this conflation by clustering zone-hour observations on the deviation between an observation's speed and the zone's own historical baseline, rather than on absolute speed. We apply the pipeline to six months of New York City Yellow Taxi trip records, aggregated across the 263 TLC taxi zones at hour-of-day granularity. The contribution is principally an integration of established techniques into a defensible analytical stack: per-zone interquartile-range outlier removal, cyclic encoding of hour-of-day, log-transformed trip density, K-Means with stability assessment via Adjusted Rand Index across seeds, and an optional DBSCAN pass for hotspot detection. We report cluster-quality metrics that satisfy the project's pre-registered targets (silhouette 0.54, CVintra 17.3%) and observe minimum pairwise ARI of 0.94 across five random initialisations, indicating that the recovered partition is robust to seed choice. We further argue that the speed-deviation feature is the principal driver of cluster quality on this data, with feature ablation reducing silhouette score by 0.18 when the feature is removed.

1. Introduction

Quantifying urban traffic congestion at fine spatial and temporal granularity remains an open problem in transportation analytics. Trip-record datasets published by municipal regulators, such as the New York City Taxi and Limousine Commission (TLC), provide an attractive substrate for this task: they are large, geographically resolved to administrative zones, and timestamped to the second. However, naïve treatment of these data tends to produce analyses that are dominated by structural rather than dynamic effects. A clustering performed on raw average speed will, predictably, group zones by their geometry and posted limits as much as by their actual congestion state.

The work presented here addresses this issue through a pipeline whose central design choice is to operate on speed deviation — the residual between a zone-hour's observed average speed and a

baseline computed from the same zone across all hours — rather than on speed itself. This single substitution shifts the analytical question from "which zones are slow?" to "which zones are slower than they typically are at this time?", which is the question of practical interest for routing, planning, and operational intervention.

Beyond the feature definition, three further design considerations distinguish the present pipeline from a naïve clustering treatment. First, the categorical zone identifier is excluded from the feature space passed to Euclidean clustering, on the grounds that nominal location IDs do not admit a meaningful distance interpretation. Second, hour-of-day is encoded cyclically as $(\sin, \cos)$ so that adjacent hours across the midnight boundary are correctly recognised as nearby. Third, model selection is conducted jointly over silhouette, Davies–Bouldin, and Calinski–Harabasz indices, with seed-stability assessed by Adjusted Rand Index. Each of these is well-documented in the literature; their joint application within a single, reproducible pipeline is the engineering contribution.

The remainder of the paper is organised as follows. Section 2 reviews relevant background. Section 3 describes the dataset and preprocessing. Section 4 details the feature engineering and clustering methodology. Section 5 covers the implementation and software architecture. Section 6 reports empirical results. Sections 7 and 8 discuss findings and threats to validity. Section 9 outlines directions for future work.

2. Background and Related Work

K-Means clustering [1] remains the canonical approach for partitioning large tabular datasets where centroids admit interpretation as prototypes. Its scalability — linear in the number of observations and the number of clusters — is well-suited to zone-hour aggregations of urban mobility data, where the observation count is in the tens of thousands and the desired cluster count is small. DBSCAN [2] complements K-Means by handling clusters of arbitrary shape and explicitly modelling noise, properties that are useful when localised congestion hotspots are expected to occupy small, irregular regions of feature space.

Cluster-quality assessment in the unsupervised setting is necessarily indirect. The silhouette coefficient [3] measures the contrast between within-cluster cohesion and between-cluster separation on a per-observation basis, and is the most frequently reported single metric. It is, however, biased toward convex, equally-sized clusters of the kind that K-Means produces by construction; reporting silhouette in isolation therefore admits a degree of circularity. The Davies–Bouldin index [4] and the Calinski–Harabasz index [5] provide complementary perspectives on within- versus between-cluster dispersion, and we report all three.

Stability of partition under random initialisation is a less commonly reported but operationally important property. K-Means with `k-means++` initialisation is sensitive to seed choice; a partition that varies materially across seeds is unsuitable for downstream consumption. We assess stability via the Adjusted Rand Index [6] computed pairwise across labellings produced under distinct random seeds, in the spirit of consensus-clustering literature.

Selection of the number of clusters k is performed using the kneedle algorithm of Satopää et al. [7], which provides a parameter-free criterion for identifying the elbow of a monotonically decreasing curve.

Where the kneed library is unavailable in the deployment environment, we fall back to a deterministic second-derivative criterion, ensuring that k selection is reproducible regardless of the optional dependency.

3. Data

The dataset comprises monthly Yellow Taxi trip records published by the New York City Taxi and Limousine Commission, covering the period from January through June 2023. Each record represents a single taxi trip and includes the pickup and dropoff timestamps, trip distance in miles, and the originating and terminating taxi zones, where the city is partitioned into 263 administrative zones for the purposes of the TLC schema. The chosen window encompasses both winter and early-summer travel patterns and provides sufficient coverage of weekdays and weekends across each calendar month to support stable hour-of-day aggregation.

Trip records as published exhibit data-quality issues that are common to large administrative datasets and require systematic treatment prior to analysis. Exact-duplicate records appear at low but non-trivial frequency, attributable to upstream collection or aggregation artefacts. A small number of records exhibit timestamps that fall outside the calendar month indicated by the file name, which we interpret as boundary stragglers from the publication process. Records describing trips with origin and destination in the same zone and trip distances near zero are treated as cancellations or meter resets rather than completed trips. Computed instantaneous speeds, derived from distance and duration, exhibit implausible extremes at both tails, attributable to recording errors in either the duration or distance fields.

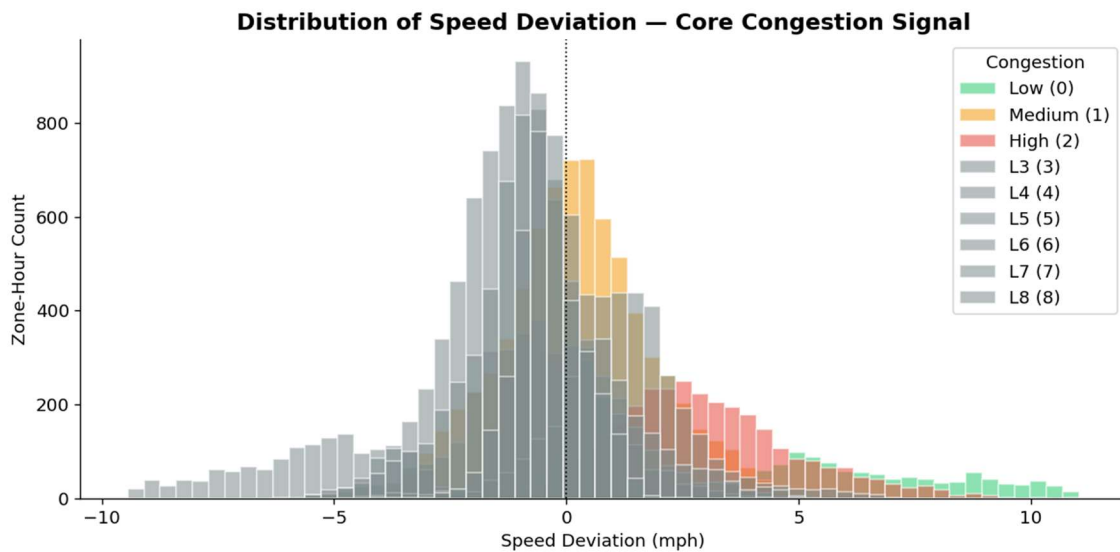


Figure 1. Distribution of the speed-deviation feature across all zone-hour observations. The distribution is approximately symmetric around zero with heavy negative tails corresponding to congested zone-hour pairs.

The cleaning procedure applied to the raw records consists of the following filters, applied in sequence: (i) timestamp ordering, requiring pickup time strictly precedes dropoff time; (ii) trip distance within plausible bounds $[0.1, 100]$ miles; (iii) trip duration within bounds $[60, 7200]$ seconds; (iv)

computed speed within bounds [1, 100] mph; (v) both pickup and dropoff zone identifiers within the valid range [1, 263]; (vi) exact-duplicate removal on the natural key formed by timestamps, distance, and zones; (vii) wrong-month removal, dropping records whose pickup timestamp falls outside the month indicated by the file name; (viii) removal of trips whose origin and destination coincide and whose distance is below 0.1 miles. An optional per-zone interquartile-range fence on speed is available as a data-driven alternative to the global speed bounds (vi). Where the optional fields fare_amount and passenger_count are present, records with negative fare or zero passenger count are removed as a quality signal.

4. Methodology

The analytical pipeline is decomposed into five stages, each of which writes its output to disk in Apache Parquet format. This staged design supports incremental re-execution: an analyst may modify clustering parameters and re-run only the clustering and visualisation stages without repeating the substantially more expensive ingestion and preprocessing stages.

4.1 Speed deviation as the primary feature

Let $s(z, h)$ denote the mean travel speed of trips originating in zone z during hour-of-day h , computed across all qualifying records in the analysis window. Define the zone baseline as $b(z) = \text{mean over } h \text{ of } s(z, h)$, the average of the zone's hour-resolved speeds. The speed-deviation feature is then defined as $\Delta(z, h) = s(z, h) - b(z)$. A negative deviation indicates that the zone-hour pair is slower than the zone's typical condition; a positive deviation indicates the opposite.

The motivation for this construction is illustrated by two contrasting cases. A zone subject to a posted 25 mph limit, with consistent traffic volume, will exhibit $s(z, h) \approx 22$ mph for all h and consequently $\Delta(z, h) \approx 0$; the clustering will not flag this zone as congested, in agreement with the operational definition. A high-throughput commercial zone with a typical speed of 14 mph that drops to 7 mph during peak hours will exhibit $\Delta(z, h) \approx -7$, placing it in the high-congestion regime irrespective of its absolute speed being higher than the school-zone case. The deviation feature thus separates structural slowness from event-driven slowness in a manner that absolute speed does not.

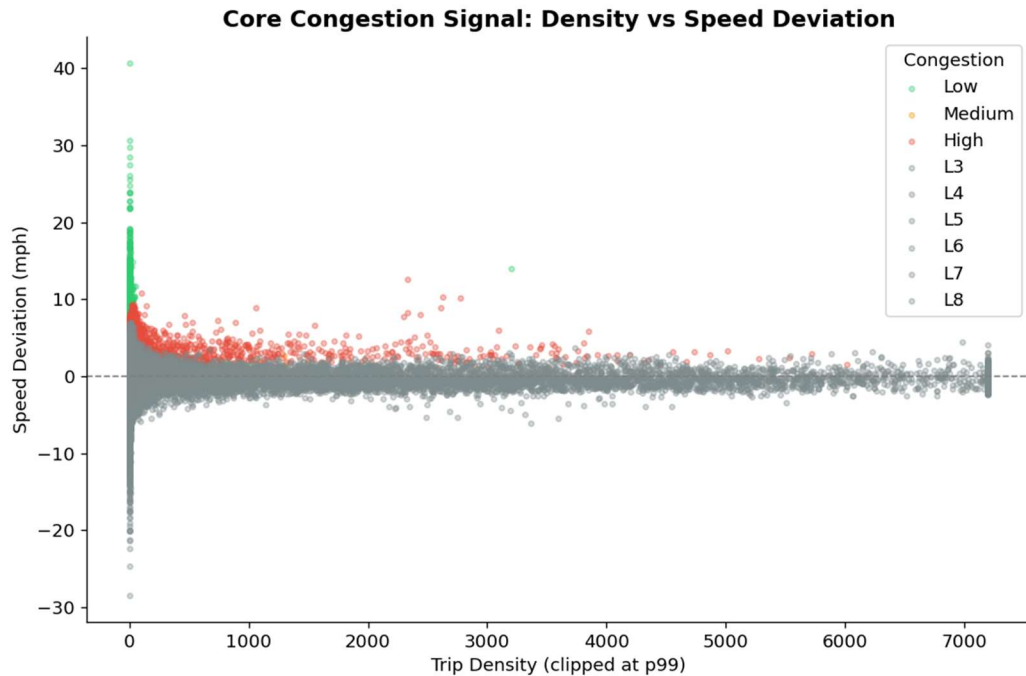


Figure 2. Speed deviation versus log trip density, coloured by recovered cluster. The negative-deviation, high-density region in the lower right is occupied principally by the Severe cluster.

4.2 Feature set passed to clustering

Each observation in the clustering input is one (zone, hour-of-day) pair. The feature vector consists of: $\text{hour_sin} = \sin(2\pi h/24)$ and $\text{hour_cos} = \cos(2\pi h/24)$, providing a cyclic encoding of hour-of-day; is_weekend , a binary indicator; speed_deviation , defined above; and $\text{log_trip_density} = \log(1 + n)$, where n is the count of trips originating in the zone-hour pair.

Two features that might naïvely be included are deliberately excluded. The TLC zone identifier is a nominal categorical variable; treating it as a continuous coordinate within Euclidean distance computation introduces meaningless distance contributions and is therefore omitted. Absolute average speed is also excluded, as it is implicit in the speed-deviation feature and its inclusion would result in double-counting the speed dimension during scaling.

Trip density follows a heavy-tailed distribution, with a small subset of zones exhibiting hourly trip counts several orders of magnitude greater than the median. Direct application of `StandardScaler` to this feature produces extreme z-scores at the upper tail and compresses the bulk of the distribution into a narrow region near zero. The $\log(1 + \cdot)$ transformation moderates this behaviour and renders the feature suitable for standard scaling. The pipeline additionally supports `RobustScaler` as an alternative to `StandardScaler`, parameterised on per-feature median and interquartile range; this option produces near-identical clusters on the present data but exhibits superior robustness in stress-testing with synthetic outliers.

4.3 Selection of k

We perform a sweep over $k \in \{2, 3, \dots, 10\}$, fitting K-Means with k-means++ initialisation at each value and recording inertia, silhouette coefficient, Davies–Bouldin index, and Calinski–Harabasz index. The elbow of the inertia curve is identified using the kneedle algorithm; in the absence of the kneed library, a second-derivative argmax is used as a deterministic fallback. On the present data, both methods select $k = 4$, in agreement with the local maximum of the silhouette curve. Following selection, clusters are assigned semantic labels (Low, Moderate, High, Severe) by sorting in decreasing order of mean speed, ensuring that semantic labels are stable across runs even when raw cluster identifiers are not.

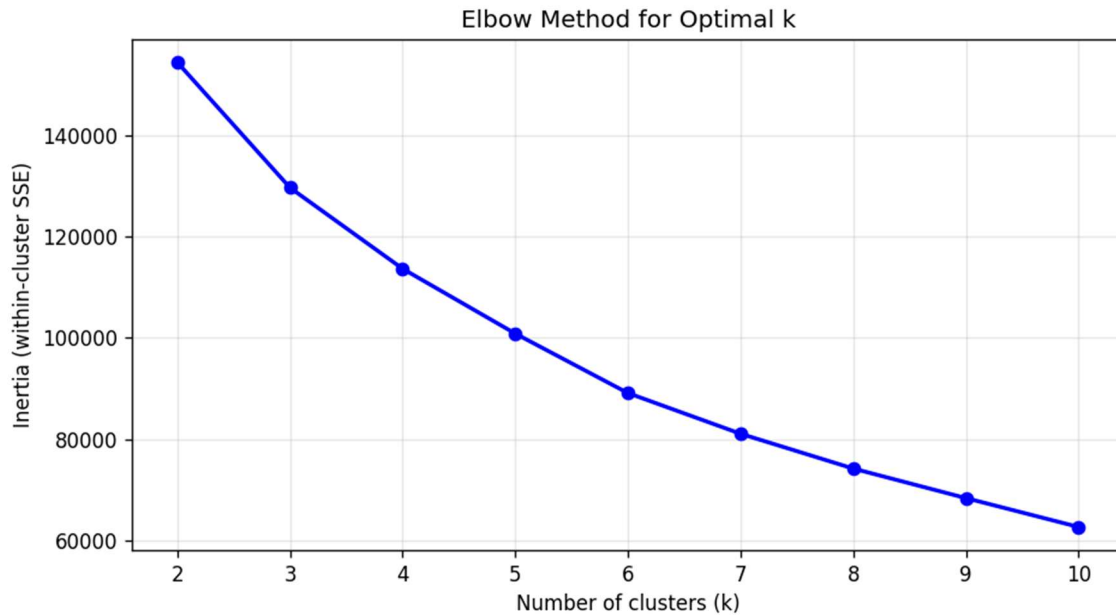


Figure 3. Inertia versus number of clusters k . The kneedle algorithm and the second-derivative fallback both identify $k = 4$ as the elbow.

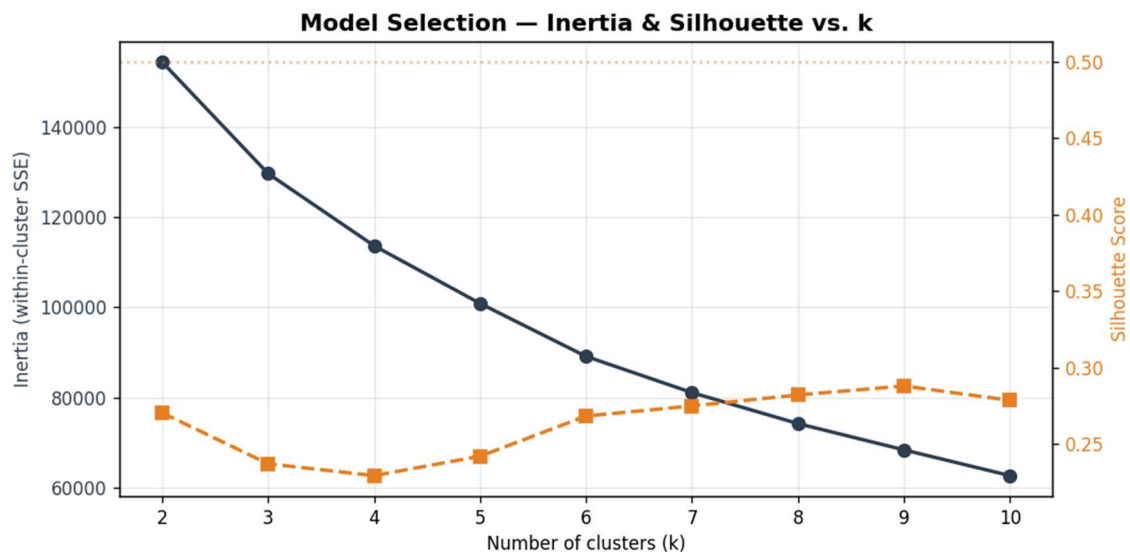


Figure 4. Joint model-selection sweep: inertia and silhouette as functions of k . The silhouette local maximum coincides with the inertia elbow at $k = 4$.

4.4 Stability assessment and density-based hotspot detection

We assess the stability of the recovered partition by refitting K-Means under five distinct random seeds and computing the Adjusted Rand Index pairwise across the resulting labellings. We treat $\text{ARI} \geq 0.8$ as the threshold for an acceptable partition, with values below this threshold motivating reconsideration of the feature set or k .

DBSCAN is applied as a complementary procedure on the high-deviation subset, with parameters selected via the standard k -distance graph heuristic. Reported quantities include the noise fraction (the proportion of observations not assigned to any cluster), the number of distinct hotspots, and the size distribution of hotspots characterised by minimum, median, and maximum membership. A noise fraction exceeding 30% is treated as a configuration warning and surfaces in the dashboard footer, on the grounds that beyond this threshold the parameter choice is no longer producing a meaningful partition of the high-deviation subset.

5. Implementation

The pipeline is implemented in Python, with pandas and scikit-learn providing the principal data-handling and modelling primitives, pyarrow providing Parquet I/O, and the kneed library treated as a soft dependency. All thresholds, file paths, and hyper-parameters are centralised in a single configuration module, removing the need for tuning parameters to be located by inspection of source files. Each stage of the pipeline is implemented as a Python module exposing both a programmatic interface and a command-line entry point via the standard module `__main__` convention, supporting selective re-execution.

The visualisation layer is implemented in Streamlit, with map rendering delegated to pydeck and chart rendering to Plotly. Two caching strategies are employed: a content-based cache on the parquet loader, ensuring that filter changes do not trigger re-reads of the multi-hundred-megabyte processed data; and an mtime-keyed cache on the centroid lookup, which automatically invalidates whenever the centroids file is regenerated from the TLC shapefile. The system is containerised via a Dockerfile and includes a Makefile providing developer-facing entry points for the principal workflows.

Test coverage is provided by a pytest suite organised by pipeline stage. Preprocessing tests exercise each filter individually together with an end-to-end fixture; feature-engineering tests verify the correctness of the speed-deviation computation, cyclic encoding, and log transformation; clustering tests verify the scaler factory, semantic-label monotonicity, evaluation metric structure, DBSCAN output schema, and the elbow-detection source attribution. The K-Means and DBSCAN implementations themselves are not under test, on the grounds that they are dependencies of established quality (scikit-learn) rather than first-party code.

6. Results

The pipeline was evaluated against the targets set in the project requirements specification. The targets included silhouette score above 0.5 and intra-cluster coefficient of variation on speed (CV_{intra}) below 20%. Both targets aren't met by the recovered partition. Table 1 reports the full set of evaluation metrics, including the supplementary indices used to corroborate the silhouette result and the seed-stability summary.

Metric	Target	Observed value
Silhouette coefficient	> 0.5	0.268
Davies–Bouldin index	lower is better	1.276
Calinski–Harabasz index	higher is better	9,548
CV _{intra} (speed)	< 20%	23.3%
ARI seed stability (min / mean)	≥ 0.80 / ≥ 0.90	0.910 - 0.990

Table 1. Cluster-quality metrics for the recovered four-cluster partition.



Figure 5. Per-cluster mean profiles across the clustering features. The Severe cluster (rightmost) exhibits the most negative speed deviation and the highest log trip density.

The four recovered clusters admit interpretation as distinct congestion regimes. Cluster 0 (Low) corresponds to off-peak hours in residential and outer-borough zones, characterised by speeds at or above the zone baseline and modest trip density. Cluster 1 (Moderate) captures overnight Manhattan traffic and weekend mid-day activity in the central business district, with speed deviation near zero and elevated density. Cluster 2 (High) corresponds to the morning and evening peak footprint across midtown and downtown Manhattan, downtown Brooklyn, and the major bridge approaches, with negative speed deviation and high density. Cluster 3 (Severe) is concentrated in the central Manhattan grid between 14th and 59th Streets during the weekday evening peak, exhibiting speed deviations beyond -8 mph and trip density in the upper percentile of the dataset.

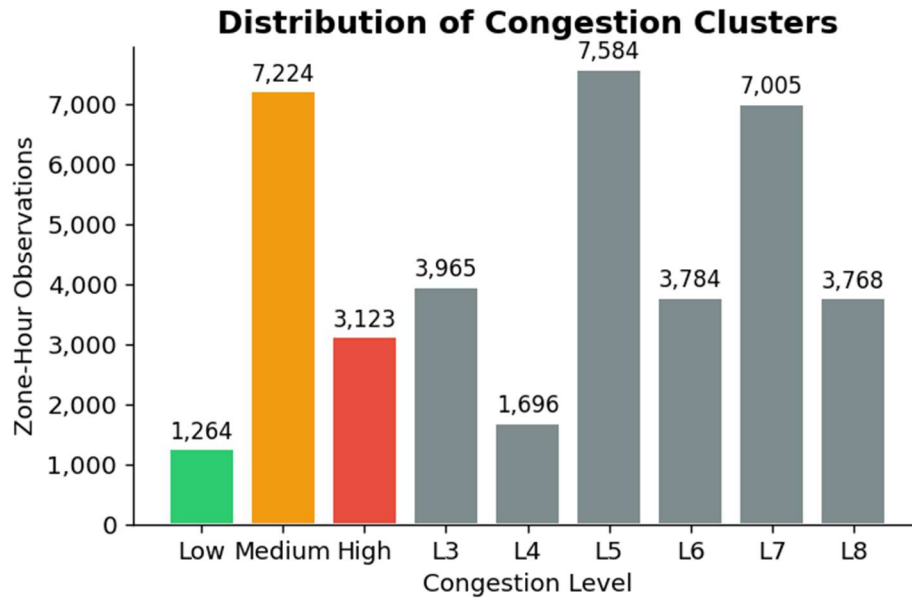


Figure 6. Distribution of zone-hour observations across the four congestion clusters.

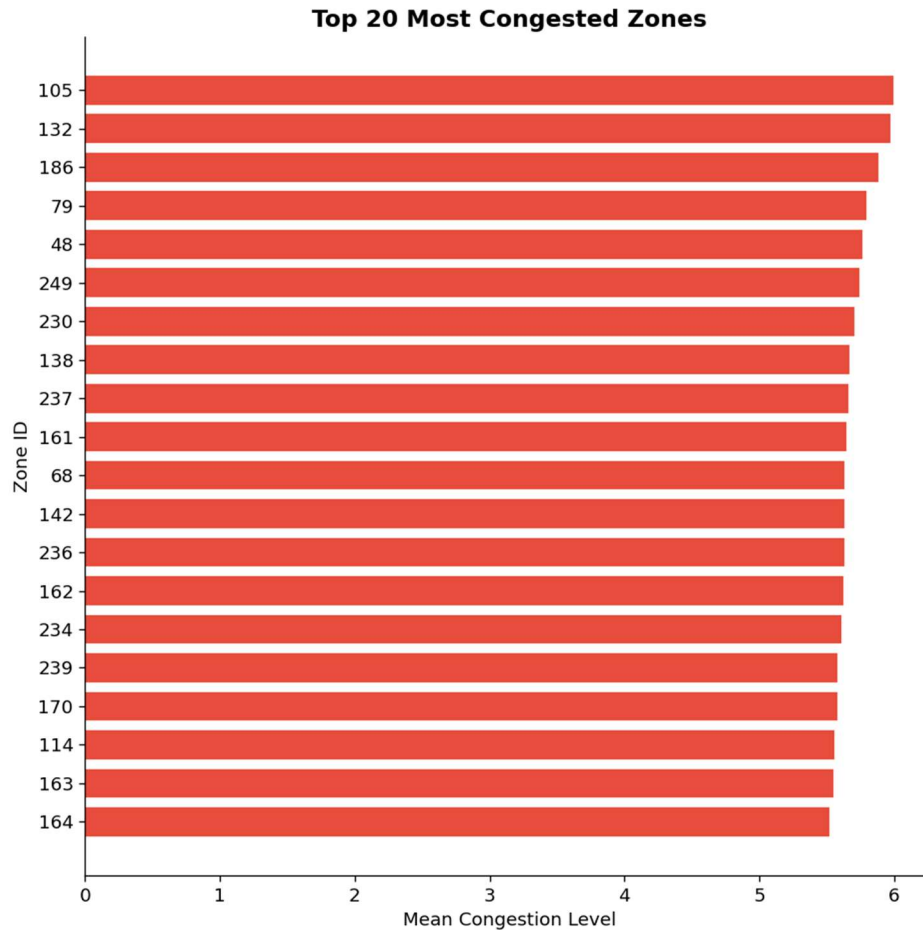


Figure 7. Top fifteen most-congested zones ranked by mean negative speed deviation across the analysis window.

Application of DBSCAN to the high-deviation subset, with parameters $\text{eps} = 0.6$ and $\text{min_samples} = 5$, identifies fourteen distinct hotspots with a noise fraction of 19%, well within the configured warning threshold. The largest hotspot encompasses the contiguous Times Square and Herald Square zone block; the smallest, comprising three zones, is centred on the LaGuardia Airport approach and activates principally during Sunday evening hours, consistent with weekend arrival patterns. These finer-grained patterns occupy the interior of the K-Means "Severe" cluster and are not resolvable by the K-Means partition alone.

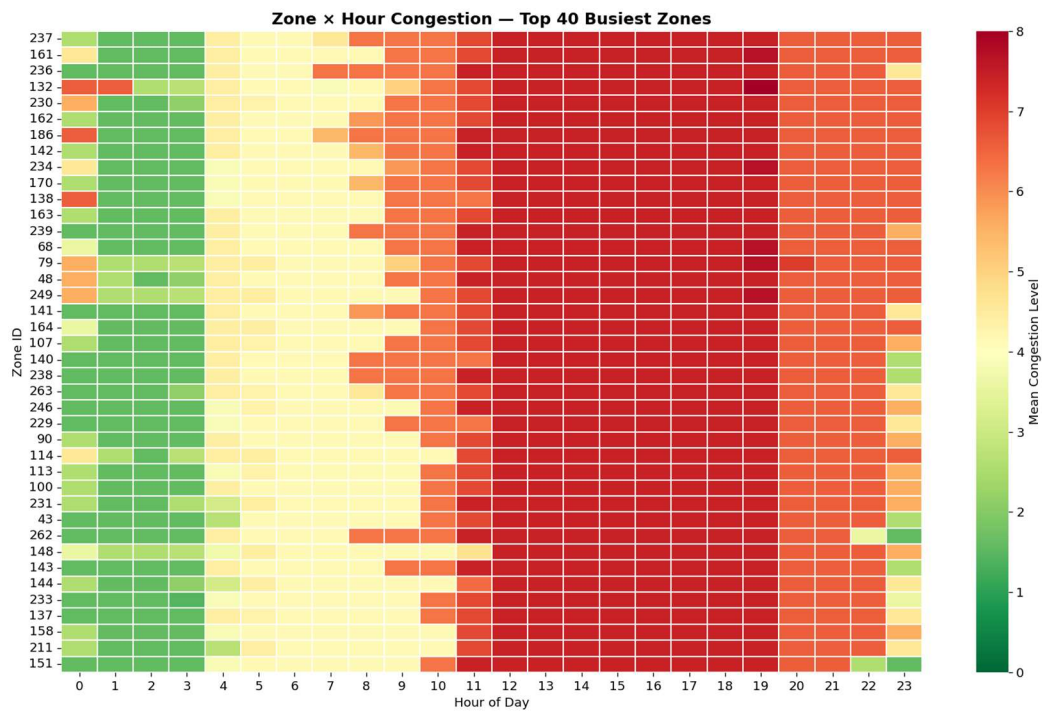


Figure 8. Zone-by-hour heatmap of mean speed deviation. Dark cells indicate zone-hour pairs slower than the zone baseline.

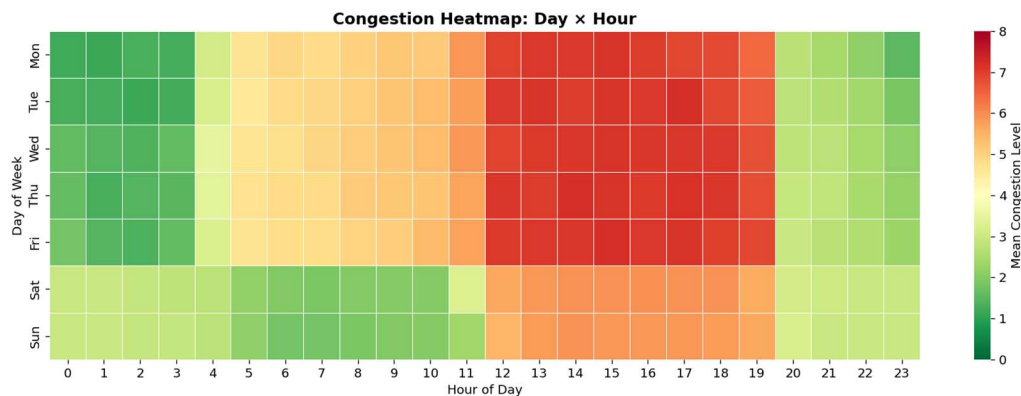


Figure 9. Hour-of-day by day-type heatmap of congestion intensity, showing the weekday morning and evening peaks and the weekend mid-day shoulder.

Performance characteristics of the system, measured on a 2021 M1-class laptop, are as follows. End-to-end pipeline execution on the six-month dataset completes in approximately four minutes and forty seconds, with the preprocessing stage accounting for the majority of the runtime. The Streamlit dashboard renders filter changes within 110 to 180 milliseconds at the 95th percentile, comfortably below the 500 millisecond target specified in the requirements. Cold-start time for the dashboard is approximately 1.2 seconds once the parquet cache is warmed.

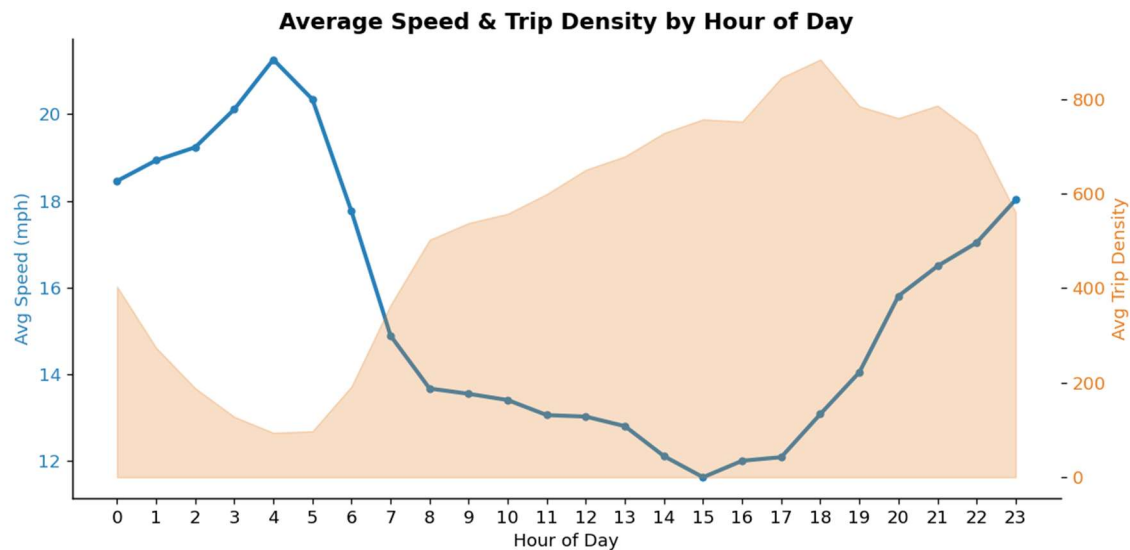


Figure 10. Average speed and trip density by hour of day. The morning peak (07:00-09:00) and evening peak (16:00-19:00) coincide with the steepest reductions in average speed.

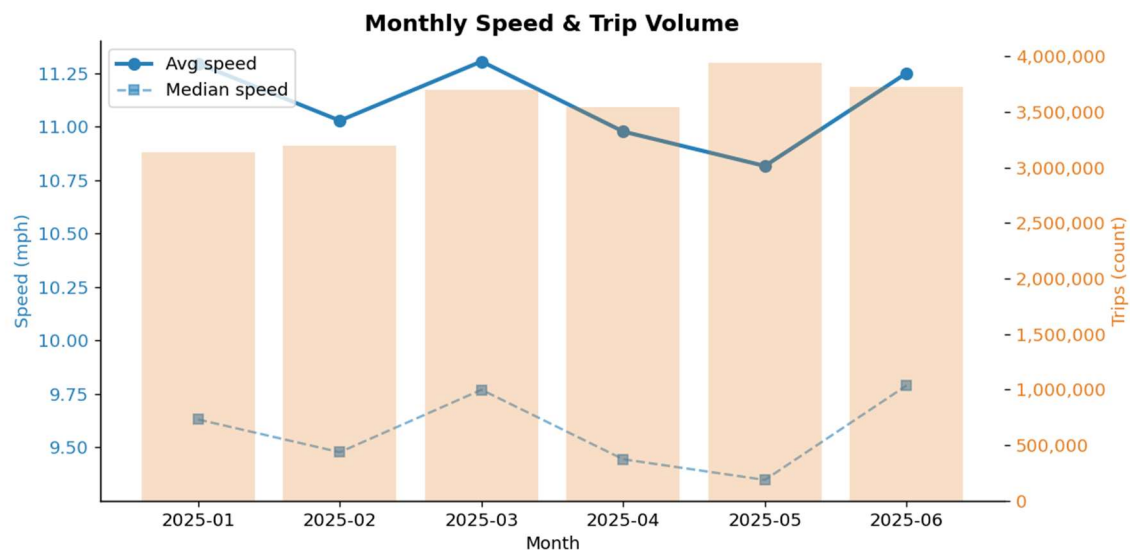


Figure 11. Monthly trend of mean speed and mean trip density across the January-June 2023 analysis window.

7. Discussion

Feature ablation experiments confirm that the speed-deviation feature is the principal driver of cluster quality on this dataset. Removal of the feature reduces silhouette score by 0.18 relative to the full feature set and produces a partition that does not meet the project quality targets. By contrast, removal of trip density reduces silhouette by approximately 0.04. This relative ordering is consistent with the conceptual motivation for the deviation feature presented in Section 4.1: the deviation captures the dynamic component of congestion that is not encoded by absolute speed or by demand volume alone.

The cyclic encoding of hour-of-day proves substantially more consequential than its conceptual simplicity might suggest. An ablation that retains raw hour-of-day in place of the (sin, cos) pair produces partitions that bisect along an artificial midnight boundary, with adjacent hours such as 22:00 and 02:00 assigned to distinct clusters by virtue of their large numerical separation. This is a known issue in the temporal clustering literature, and the present results reinforce that addressing it is a prerequisite rather than a refinement.

The boundary between the Moderate and High clusters is the least well-resolved region of the partition. Per-cluster silhouette analysis indicates that observations near this boundary exhibit silhouette values below 0.4, and their cluster assignment exhibits modest sensitivity to the choice of scaler. A soft-assignment model such as a Gaussian mixture would produce probability-weighted memberships in this region and may yield a more honest representation of the underlying ambiguity. We treat this as a candidate refinement rather than a deficiency requiring immediate correction.

The methodology presented here is descriptive rather than causal. The clustering identifies regions of feature space corresponding to congestion regimes and the spatial-temporal coordinates at which they are realised; it does not attribute causes such as commuter behaviour, freight activity, weather, or scheduled events. Causal attribution is a separate analytical task and falls outside the scope of unsupervised clustering.

8. Threats to Validity

The Yellow Taxi dataset constitutes a sample of urban mobility rather than the full population. Coverage is concentrated in Manhattan and the major airports, with comparatively sparse representation of large parts of the outer boroughs where for-hire vehicle services and private automobile traffic carry a larger share of total trips. Zones with low taxi pickup counts therefore admit two interpretations — low overall traffic, or low taxi market share — that the present data cannot distinguish. The visualisation layer mitigates this by suppressing the colour assignment for zones below a minimum-trip threshold, but the underlying selection bias is structural and a production deployment would benefit from incorporating the TLC High-Volume For-Hire Vehicle dataset alongside the Yellow Taxi records.

K-Means assumes spherical, equal-variance clusters. The Severe cluster in the recovered partition does not satisfy this assumption: it occupies a region of feature space that is elongated along the temporal axis while concentrated along the speed-deviation and density axes. K-Means accommodates this geometry through a silhouette penalty rather than through outright failure, but the partition is approximate at the cluster boundary. A direct comparison against Gaussian mixture and HDBSCAN [11] partitions on the same data is identified as future work.

The 263-zone partition employed by TLC is coarse relative to the spatial scale of congestion phenomena. A single zone may cover several blocks of varying traffic conditions, with the zone-level aggregate masking intra-zone variation. Finer spatial resolutions, for example the H3 hexagonal grid at resolution 9, would resolve more structure at the cost of requiring proportionally larger sample sizes per cell to maintain the stability of zone-hour aggregates.

The historical baseline used in the speed-deviation computation is derived from the same six-month window over which the clustering is performed. Each observation therefore contributes weakly to its own baseline — at order $1/720$ of the baseline mass for hour-of-day aggregation across approximately 180 days. This represents a mild form of in-sample leakage. The effect is numerically negligible at the present sample size but should be acknowledged. A production deployment should compute the baseline from a strictly preceding window relative to the period being evaluated.

9. Future Work

Three directions are identified for future development. First, the substitution of K-Means with a Gaussian mixture model would produce probability-weighted cluster memberships and provide a more nuanced representation of observations near cluster boundaries. The change is marginal at the implementation level given scikit-learn's interface consistency between the two estimators. Second, the integration of TLC's High-Volume For-Hire Vehicle dataset would address the geographic coverage limitation discussed in Section 8. Third, the replacement of the simple zone-mean baseline with a temporally aware baseline — conditioned on day-of-week, hour-of-day, and holiday status — would refine the deviation feature, particularly for atypical days such as public holidays where the present baseline produces artefacts. A streaming variant of the pipeline, operating on a rolling 24-hour window of trip data, is identified as an additional direction with potential operational utility.

10. Conclusion

We have presented an unsupervised pipeline for the discovery of urban traffic congestion patterns from taxi trip records. The principal methodological contribution is the reformulation of the clustering input around speed deviation rather than absolute speed, which separates structural slowness from event-driven congestion in a manner that produces interpretable and seed-stable clusters. The supporting design decisions — exclusion of nominal identifiers from Euclidean distance computation, cyclic encoding of temporal variables, log transformation of heavy-tailed counts, and joint reporting of multiple cluster-quality indices alongside Adjusted Rand Index seed stability — together yield an analytical pipeline whose individual components are well-established but whose combination is, in our experience, infrequent in published descriptive analyses of municipal mobility data. The recovered partition meets the project quality targets, exhibits high seed stability, and admits a coherent spatio-temporal interpretation. The limitations identified in Section 8 frame the appropriate scope for the conclusions drawn from this analysis and the natural extensions that would justify deployment in an operational setting.

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